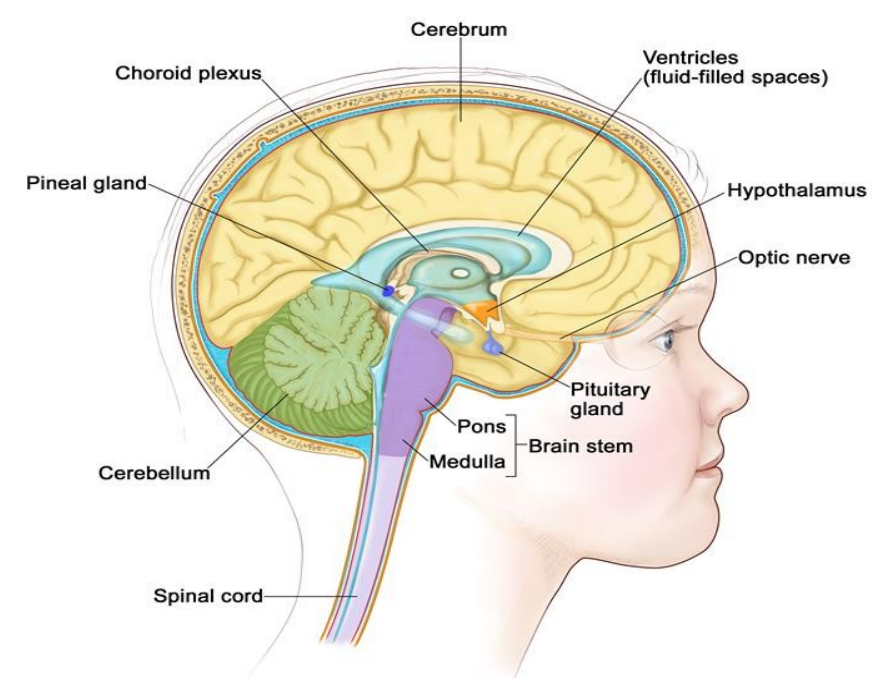
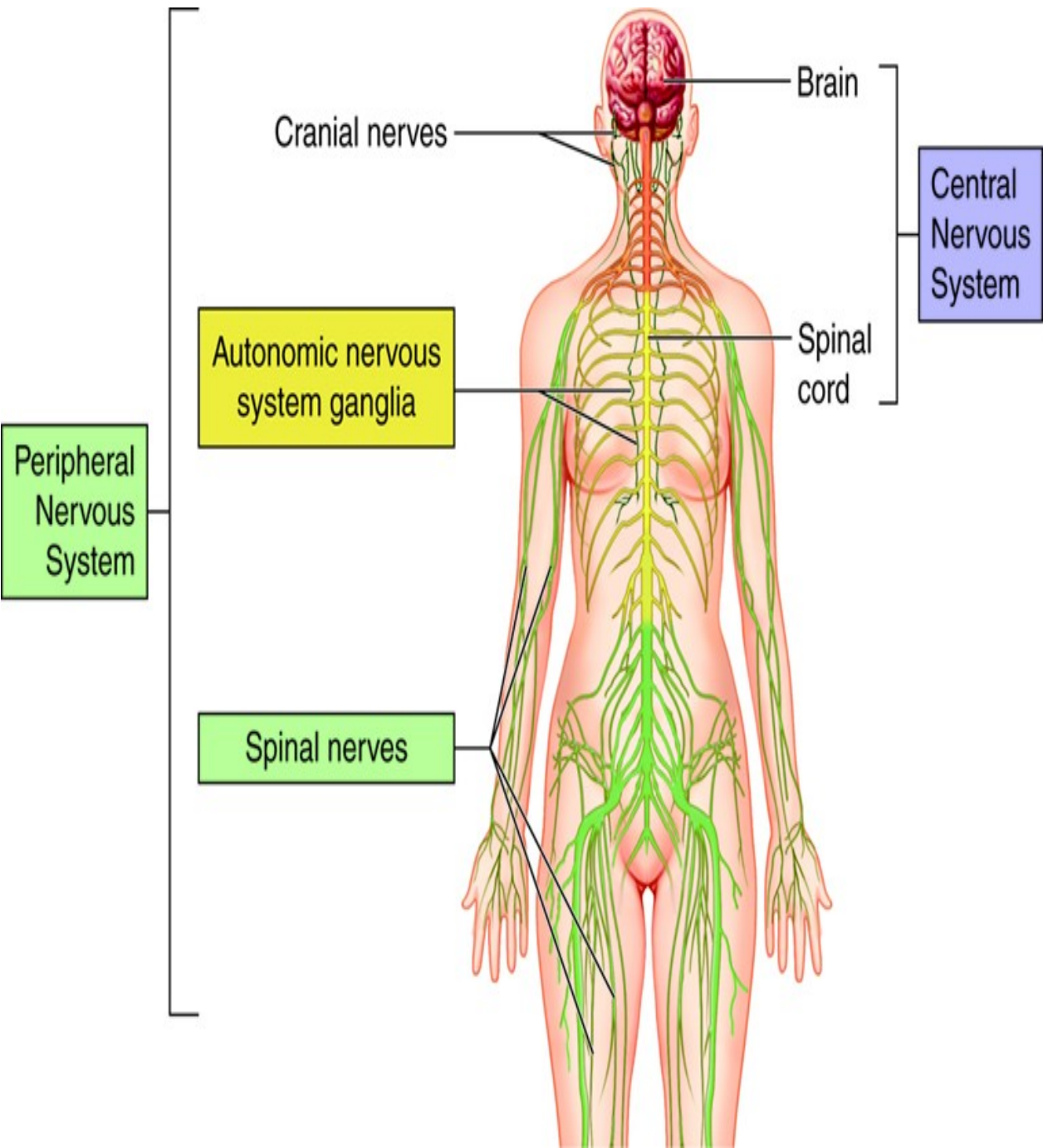


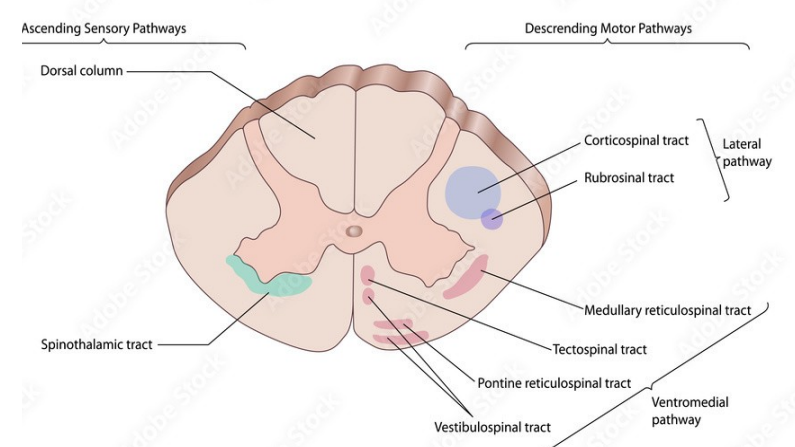
Neurologic examination

Objectives

- To learn the essential components of a neurological examination.
- To connect clinical findings with relevant pharmacological treatments and follow up.



THE SPINAL CORD
Cross-Sectional Anatomy



Nervous System

General Considerations

Always consider left to right **symmetry**

It includes:

- Mental Status

- Cranial Nerves

- Motor

- Coordination and gait

- Reflexes

- Sensory

Mini Mental Status Examination

- 1. Orientation in time, place, person
- 2. Registration
- 3. Attention and calculation
- 4. Recall
- 5. Language
 - Total score out of 30_____

The Glasgow Coma Scale

Glasgow Coma Scale		
Response	Scale	Score
Eye Opening Response	Eyes open spontaneously	4 Points
	Eyes open to verbal command, speech, or shout	3 Points
	Eyes open to pain (not applied to face)	2 Points
	No eye opening	1 Point
Verbal Response	Oriented	5 Points
	Confused conversation, but able to answer questions	4 Points
	Inappropriate responses, words discernible	3 Points
	Incomprehensible sounds or speech	2 Points
	No verbal response	1 Point
Motor Response	Obeys commands for movement	6 Points
	Purposeful movement to painful stimulus	5 Points
	Withdraws from pain	4 Points
	Abnormal (spastic) flexion, decorticate posture	3 Points
	Extensor (rigid) response, decerebrate posture	2 Points
	No motor response	1 Point
Minor Brain Injury = 13-15 points; Moderate Brain Injury = 9-12 points; Severe Brain Injury = 3-8 points		

Cranial Nerves

I – Olfactory

II – Optic

III – Oculomotor

IV – Trochlear

V – Trigeminal

VI – Abducent

VII – Facial

VIII – Vestibulocochlear

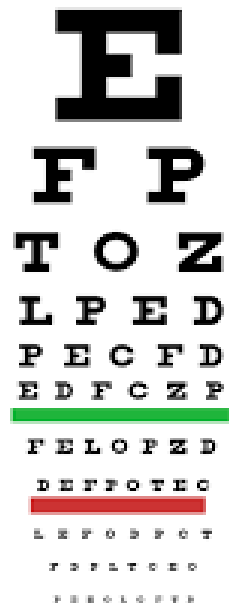
IX – Glossopharyngeal

X – Vagus

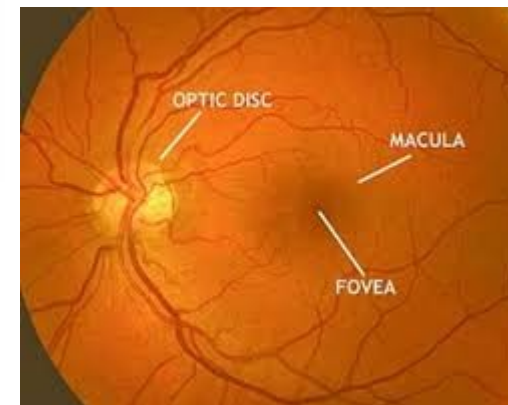
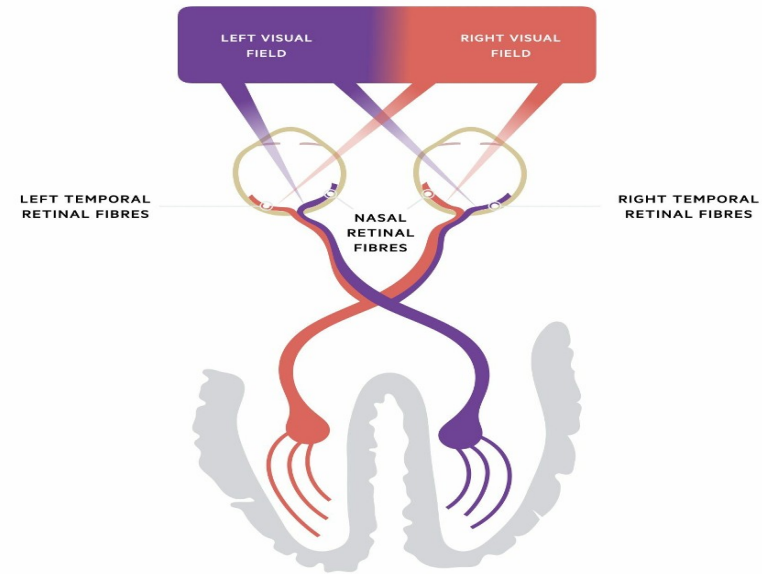
XI – Accessory

XII - Hypoglossal

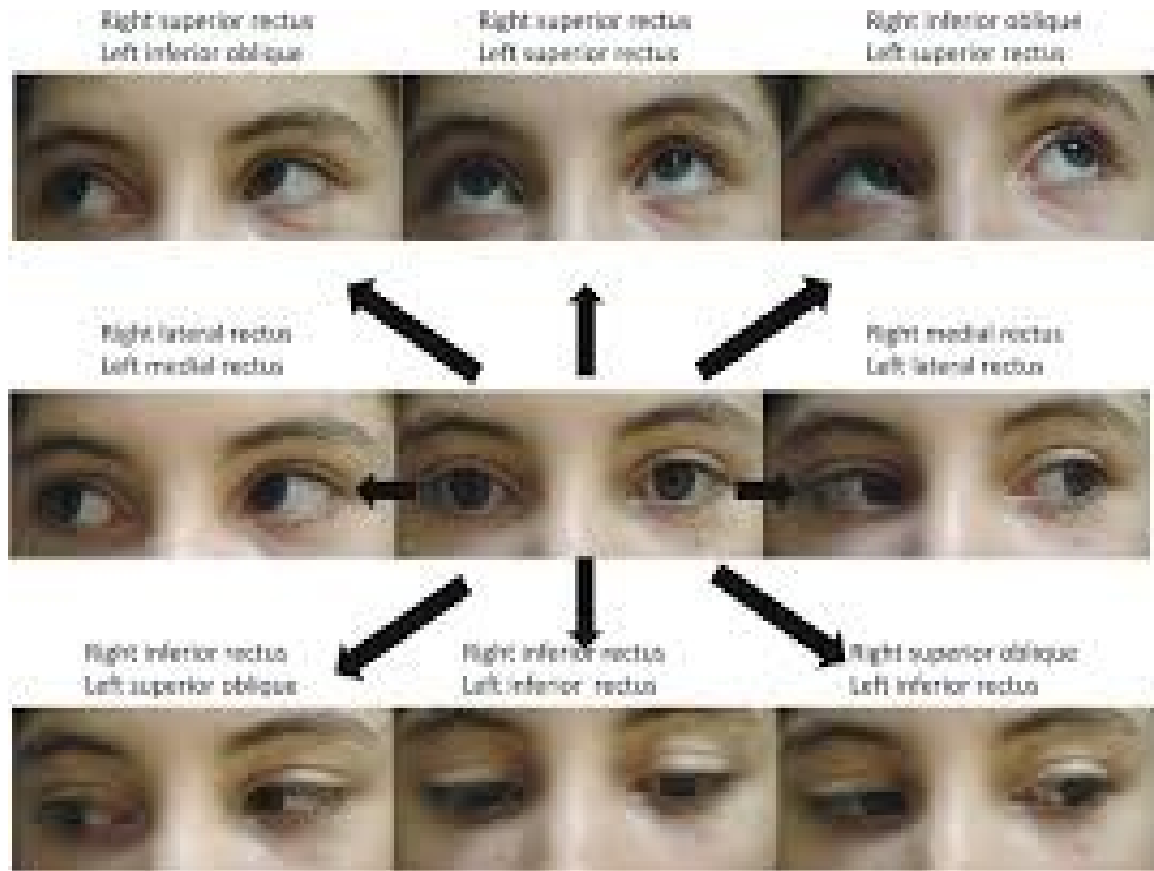
CN II



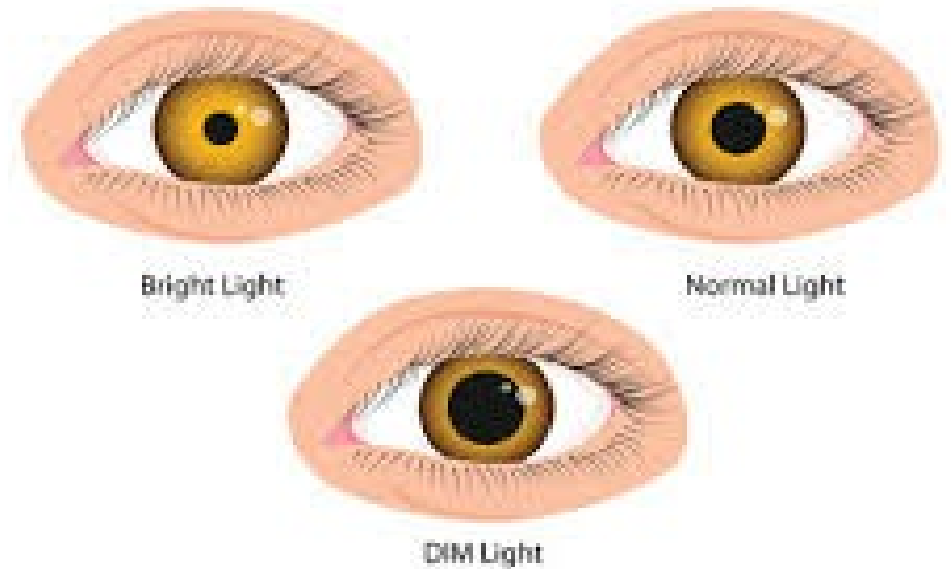
1	20/200
2	20/100
3	20/70
4	20/50
5	20/40
6	20/30
7	20/25
8	20/20
9	
10	
11	



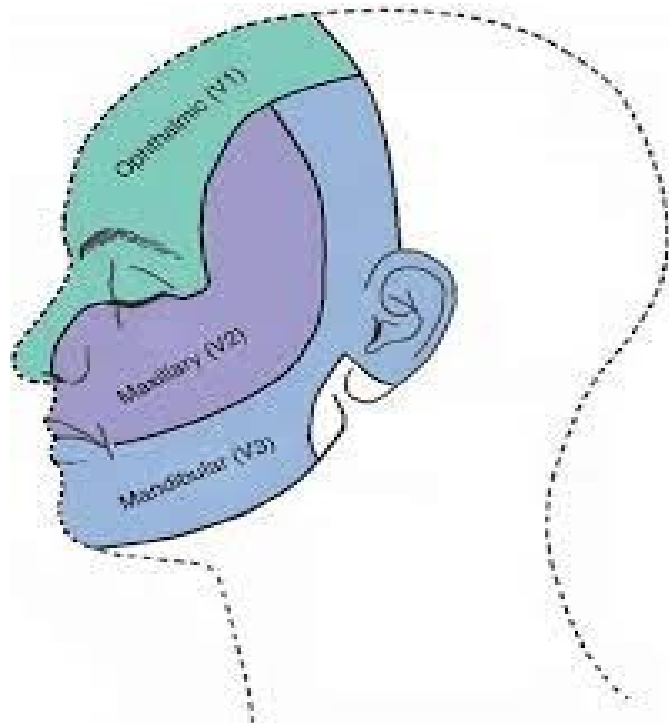
CN III, IV, VI



PUPILLARY LIGHT REFLEX



CN V



CN VII



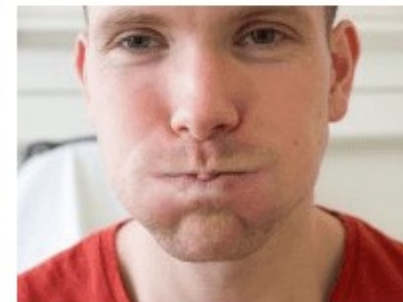
Crease up the forehead



Keep eyes closed against resistance



Reveal the teeth



Puff out the cheeks

FACIAL NERVE PALSY

Inability to wrinkle brow:

The affected person may not be able to raise one eyebrow or wrinkle the forehead on one side.

Drooping eyelid, inability to close eye:

The eyelid on one side may droop, making it difficult to close the eye completely.

Asymmetrical smile:

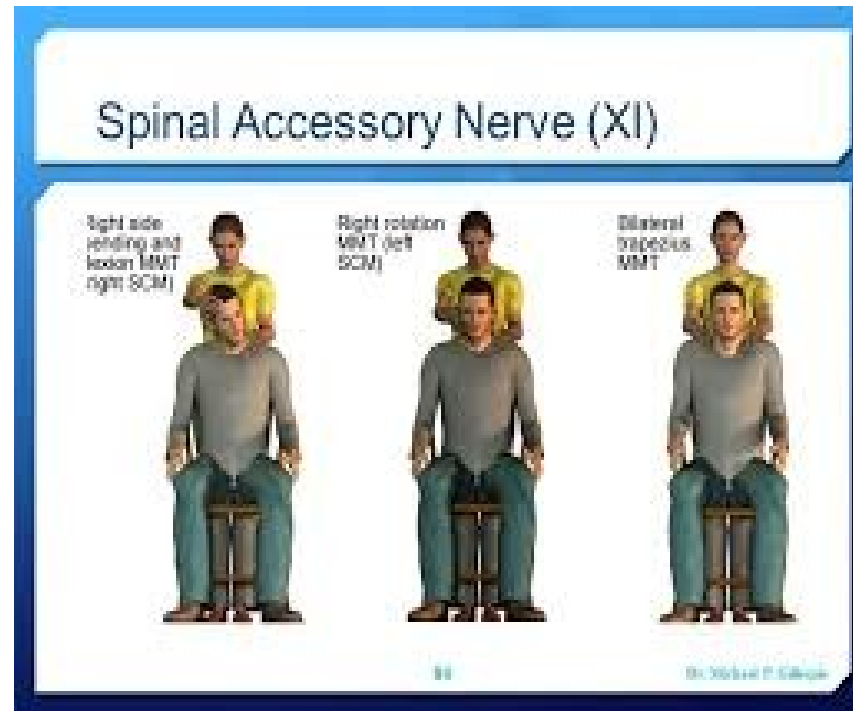
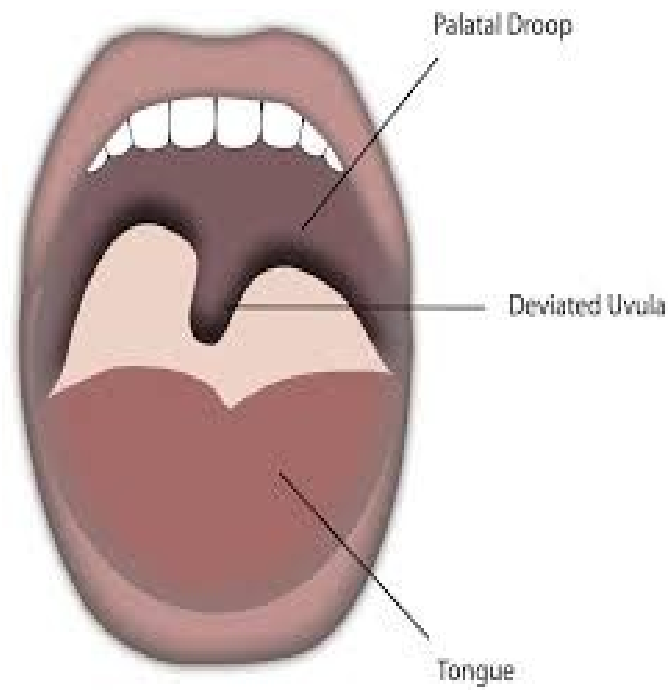
When the person smiles, the affected side of the face may not move as much as the other side, resulting in an uneven or lopsided smile.

Drooping corner of mouth:

The corner of the mouth on one side may droop, causing a lopsided appearance.

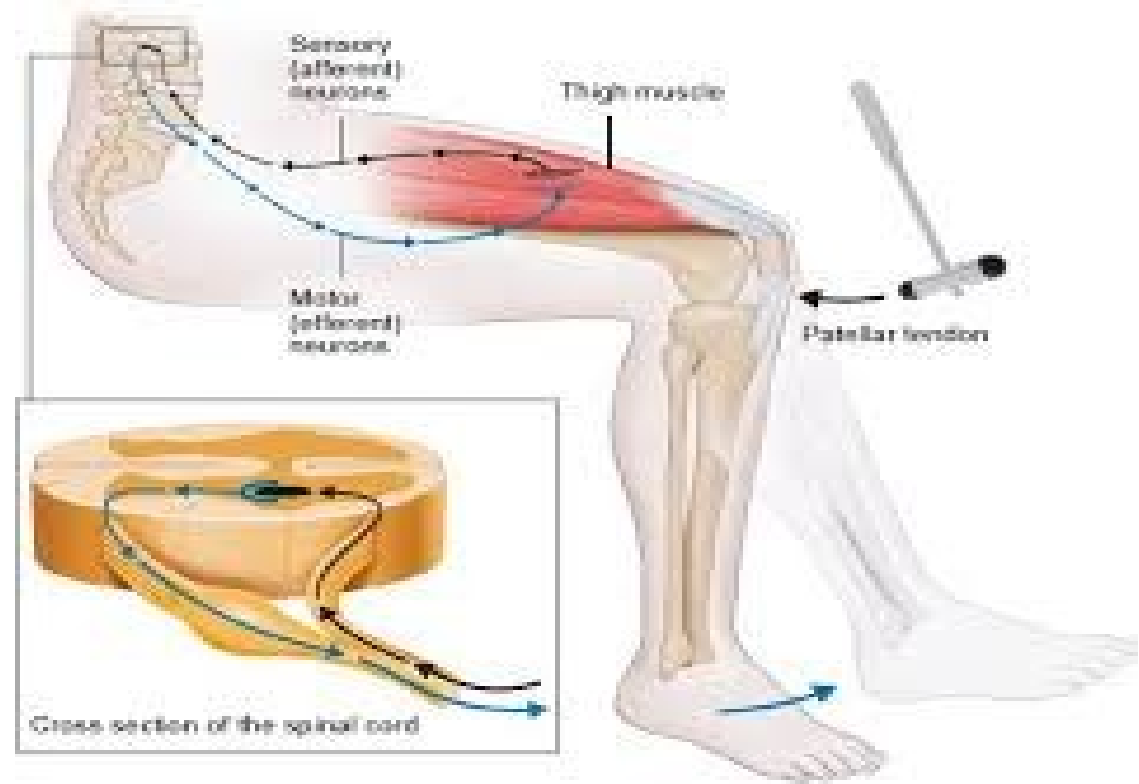


Cranial Nerves



Motor Examination

- Symmetry
- Bulk
- Tone
- Power
- Reflex



Grading Motor Strength

Grade	Description
0/5	No muscle movement
1/5	Visible flicker muscle movement, but no movement at the joint
2/5	Movement at the joint, but not against gravity
3/5	Movement against gravity, but not against added resistance
4/5	Movement against resistance, but less than normal
5/5	Normal strength

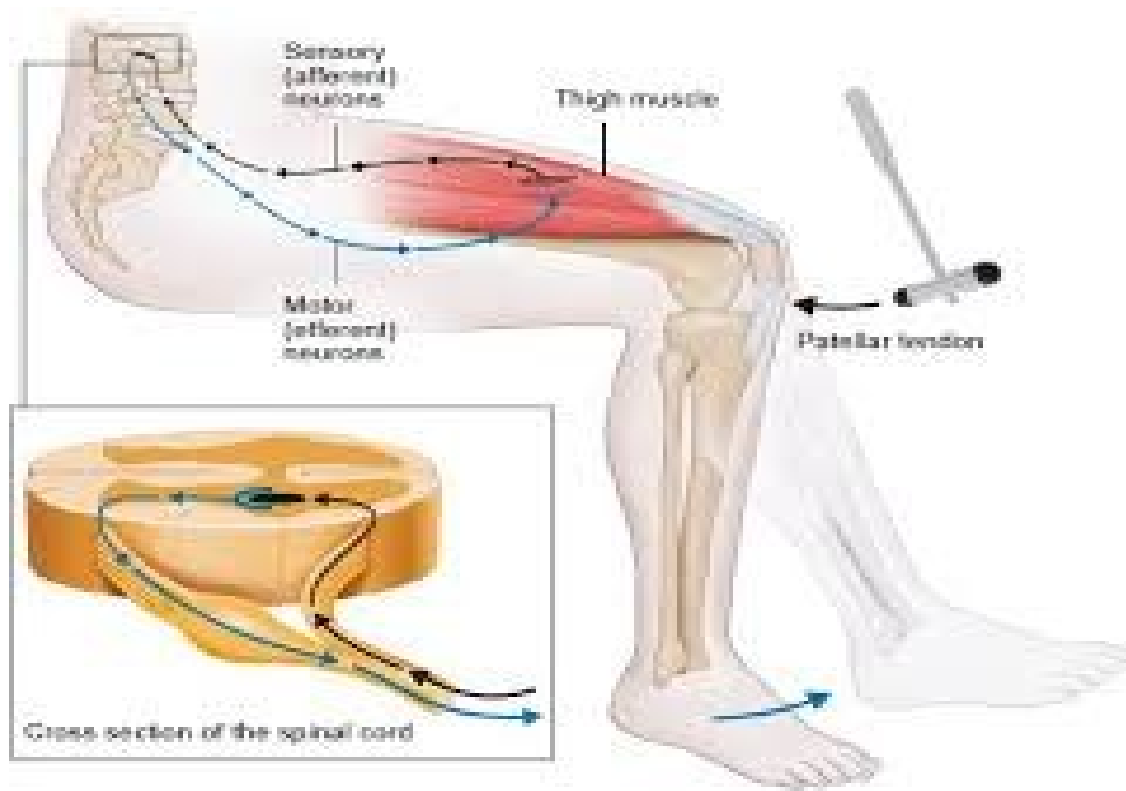


Table 8.7 Grading scale of deep tendon reflex

Deep Tendon Reflex Grading Scale

Grade	Description
0	Absent (with Jendrassik maneuver)
1+ or +	Hypoactive (less brisk)
2+ or ++	"Normal" (brisk)
3+ or +++	Hyperactive without clonus (very brisk)
4+ or ++++	Hyperactive with clonus

Differentiate between upper and lower motor neuron lesions

	UMNL	LMNL
Lesion	Above the anterior horn cell in the spinal cord or above the nuclei of the cranial nerves	Anterior horn cell, motor nerve fibre or neuromuscular junction
Tone	Increased (spasticity) $\pm$ clonus	Reduced
Muscle weakness	All muscle groups of the lower limb – more marked in the flexor muscles. In the upper limb weakness is more marked in the extensors	More distally than proximally. Both flexors and extensors affected
Deep tendon reflexes	Increased (but superficial reflexes such as abdominal reflexes are usually absent)	Reduced or absent
Plantar response	Extensor (upgoing toe)	Normal or absent
Fasciculation	Absent	May be present in anterior horn cell lesions
Wasting	Late; mainly because of disuse	Usually present

TABLE 183-2 Findings Helpful for Localizations within the Nervous System (Continued)

	SIGNS
Brainstem	Isolated cranial nerve abnormalities (single or multiple) “Crossed” weakness ^a and sensory abnormalities of head and limbs, e.g., weakness of right face and left arm and leg
Spinal cord	Back pain or tenderness Weakness ^a and sensory abnormalities sparing the head Mixed upper and lower motor neuron findings Sensory level Sphincter dysfunction
Spinal roots	Radiating limb pain Weakness ^b or sensory abnormalities following root distribution (see Figs. 183-1 and 183-2) Loss of reflexes
Peripheral nerve	Mid or distal limb pain Weakness ^b or sensory abnormalities following nerve distribution (see Figs. 183-1 and 183-2) “Stocking or glove” distribution of sensory loss Loss of reflexes
Neuromuscular junction	Bilateral weakness including face (ptosis, diplopia, dysphagia) and proximal limbs Increasing weakness with exertion Sparing of sensation
Muscle	Bilateral proximal or distal weakness Sparing of sensation

^aWeakness along with other abnormalities having an “upper motor neuron” pattern, i.e., spasticity, weakness of extensors > flexors in the upper extremity and flexors > extensors in the lower extremity, and hyperreflexia.

^bWeakness along with other abnormalities having a “lower motor neuron” pattern, i.e., flaccidity and hyporeflexia.

TABLE 55-2 Common Causes of Weakness

Upper Motor Neuron

Cortex: Ischemia; hemorrhage; intrinsic mass lesion (primary or metastatic cancer, abscess); extrinsic mass lesion (subdural hematoma); degenerative (amyotrophic lateral sclerosis)

Subcortical white matter/internal capsule: Ischemia; hemorrhage; intrinsic mass lesion (primary or metastatic cancer, abscess); immunologic (multiple sclerosis); infectious (progressive multifocal leukoencephalopathy)

Brainstem: Ischemia, immunologic (multiple sclerosis)

Spinal cord: Extrinsic compression (cervical spondylosis, metastatic cancer, epidural abscess); immunologic (multiple sclerosis, transverse myelitis); infectious (AIDS-associated myelopathy, HTLV-I-associated myelopathy, tabes dorsalis); nutritional deficiency (subacute combined degeneration)

Motor Unit

Spinal motor neuron: Degenerative (amyotrophic lateral sclerosis), infectious (poliomyelitis)

Spinal root: Compressive (degenerative disk disease); immunologic (Guillain-Barré syndrome); infectious (AIDS-associated polyradiculopathy, Lyme disease)

Peripheral nerve: Metabolic (diabetes mellitus, uremia, porphyria); toxic (ethanol, heavy metals, many drugs, diphtheria); nutritional (B_{12} deficiency); inflammatory (polyarteritis nodosa); hereditary (Charcot-Marie-Tooth); immunologic (paraneoplastic, paraproteinemia); infectious (AIDS-associated polyneuropathies and mononeuritis multiplex); compressive (entrapment)

Neuromuscular junction: Immunologic (myasthenia gravis); toxic (botulism, aminoglycosides)

Muscle: Inflammatory (polymyositis, inclusion body myositis); degenerative (muscular dystrophy); toxic (glucocorticoids, ethanol, AZT); infectious (trichinosis); metabolic (hypothyroid, periodic paralyses); congenital (central core disease)

Sensory examination

- Vibration
- Light Touch
- Position Sense
- Pain
- Temperature
- Light Touch
- Discrimination

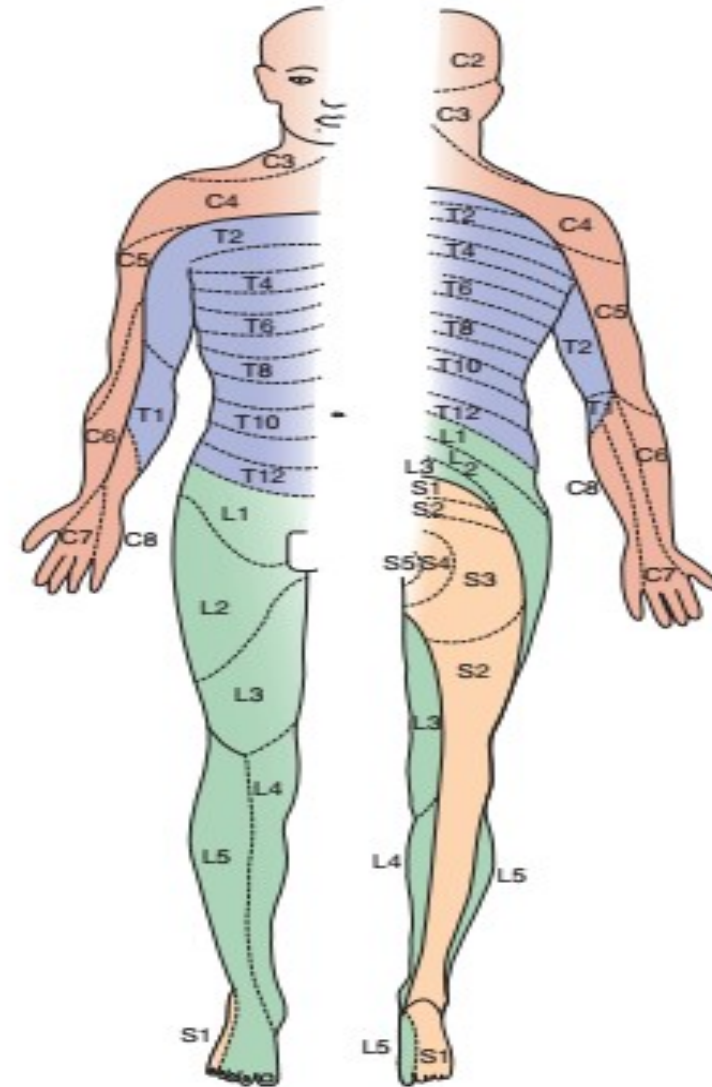


FIGURE 183-2 Distribution of the sensory spinal roots on the surface of the body (dermatomes). (From D Sinclair: *Mechanisms of Cutaneous Sensation*. Oxford, UK, Oxford University Press, 1981; with permission from Dr. David Sinclair.)

- Coordination Tests



Meningeal signs

